

Linear systems and elimination

Near-quadratic solution cost at a prescribed backward error

Difficulty: extreme **Importance:** broadly interesting

Status: open; removing dimension-dependent overhead from general backward-error solution cost.

Problem statement

Does there exist a randomized algorithm and absolute constants $C, q > 0$ such that, for every $n \geq 1$, nonsingular $A \in \mathbb{R}^{n \times n}$ with $\|A\|_2 = 1$, $b \neq 0$, and $0 < \varepsilon < 1/2$, it returns $x \neq 0$ satisfying

$$\Pr \left\{ \frac{\|Ax - b\|_2}{\|A\|_2 \|x\|_2} \leq \varepsilon \right\} \geq 0.99$$

using at most $Cn^2\varepsilon^{-q}$ operations? Use an exact-real arithmetic model with scalar arithmetic, square roots, comparisons, and independent standard Gaussian or random-bit draws at unit cost. All input entries may be accessed; the norm normalization is an input promise. The cost constants must be independent of the condition number and of n, b, ε . The fraction is normwise backward error with perturbations allowed in A only. A deterministic algorithm meeting the bound would also answer positively.

References and status

M. Dereziński, Y. Nakatsukasa, and E. Rebrova, *Towards Universal Convergence of Backward Error in Linear System Solvers* (22 May 2026), §§1, 3.1, 5.2, 7, poses the cost question and specifies the backward-error metric. Corollary 17 gives $O(n^2/\sqrt{\varepsilon})$ for PSD systems. The revised paper's Corollary 25 proves the general-system bound $O(n^2 \log(n/\delta)/\varepsilon)$ with failure probability δ ; its §7 asks for convergence independent of dimension. The remaining target here removes the $\log n$ overhead, which cannot be hidden in a constant depending only on ε . The model and success probability above make this explicit. Searches on 2026-09-08 for “universal backward error 2026 linear systems”, “MINBERR smoothed analysis”, and “quadratic backward error linear solver” found no resolution of the displayed bound. The earlier version's lack of a proved general-system rate is no longer current.